

1. **Sum of Numbers**

Write a program that calculates the sum of all numbers from 1 to 100 using a loop.

2. **Multiplication Table**

Write a program that prints the multiplication table for a given number (e.g., 5) up to 10.

3. **FizzBuzz**

Write a program that prints numbers from 1 to 100. For multiples of 3, print "Fizz"; for multiples of 5, print "Buzz"; and for multiples of both, print "FizzBuzz".

4. **Factorial Calculator**

Write a function that calculates the factorial of a given number. (e.g., $5! = 5 * 4 * 3 * 2 * 1 = 120$).

5. **Reverse a String**

Write a function that takes a string and returns the reversed version of it.

6. **Palindrome Checker**

Write a function that checks if a given string is a palindrome (e.g., "madam" or "racecar").

7. **Sum of Array Elements**

Write a function that takes an array of numbers and returns the sum of all elements.

8. **Find the Maximum**

Write a function that takes an array of numbers and returns the largest number in the array.

9. **Remove Duplicates**

Write a function that takes an array and returns a new array with duplicate values removed.

Bonus Challenges

- **Count Vowels**

Write a function that counts the number of vowels in a given string.

- **Sort an Array**

Write a function that sorts an array of numbers in ascending order.

- **Object Manipulation**

Create an object representing a person with properties like name, age, and city. Write a function to update the person's age.